

# Varun Srivastava

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## EDUCATION

**University of California, Berkeley**

*B.A. Computer Science, Mathematics*

Graduated Dec 2023

GPA: 3.9/4

## EXPERIENCE

**Machine Learning Engineer**

*AdMarketplace*

August 2024 – Present

New York, NY

- Lead the training and deployment of the embedding (vector) model powering ad retrieval across all major publishers (such as Firefox and Klarna), serving 100M+ product titles and driving > 100M in annual revenue.
- Train and optimize CTR/CPC time-series models for the firm's algorithmic bidding system, directly impacting millions of dollars in daily ad spend allocation.
- Built an AI shopping assistant app which is used to demo real-time intent-based ad retrieval to prospective clients.

**Founding Engineer**

*Stork Oracle Network*

August 2022 – December 2022

San Francisco, CA

- Designed a high-speed and reliable WebSocket API for live crypto data. The pricefeed served several notable paying DeFi clients, such as Vertex and Apex, which traded millions of dollars daily.

**Quantitative Analysis Intern**

*Dexterity Capital*

May 2022 – August 2022

Seattle, WA

- Implemented an evolutionary hyperparameter tuning model for the company's proprietary HFT market making algorithm, trading billions in daily volume; improving search time by > 100x.

## PROJECTS

**NanoGPT Speedrun / NanoChat** | [github.com/KellerJordan/modded-nanogpt](https://github.com/KellerJordan/modded-nanogpt) | [github.com/karpathy/nanochat](https://github.com/karpathy/nanochat)

- Top three competitor in a popular LLM training competition (Modded NanoGPT), with several records involving novel pretraining algorithmic research.
- Notable contributions to NanoChat (> 40,000 Github Stars), an affordable end-to-end GPT2 reproduction inspired by the NanoGPT competition.

**Spotify MCP** | [github.com/varunneal/spotify-mcp](https://github.com/varunneal/spotify-mcp)

- Built an MCP Server that allows LLMs to control Spotify. Winner at the Anthropic MCP Hackathon (Nov 2024).

**Semantic Finder** | [do-me.github.io/SemanticFinder](https://do-me.github.io/SemanticFinder)

- Developed one of the first in-browser vector search engines and a corresponding Chrome extension (300 Stars).

## RESEARCH & PUBLICATIONS

Machine Learning & Vision Researcher | *Berkeley Visual Computing Lab*

December 2023 – May 2024

- Lee, J., **Srivastava, V.**, Jennings, N., Ng, R. "Theory of Human Tetrachromatic Color Experience." *ACM SIGGRAPH*, 2024. Honorable Mention for Technical Papers Awards, spotlighted at the annual conference.
- Researched the emergence of color vision in humans via neurological models, specifically in the case of esoteric conditions such as human tetrachromacy.

Vision & Neuroscience Researcher | *Na-Ji Biophysics Lab*

September 2019 – April 2020

- Implemented computer vision tools in Matlab for researchers to track the pupil size of rats in video data.
- Correlated terabytes of time-series data with pupil measurements to analyze how rodents react to shock stimulus.

## TECHNICAL SKILLS

**Languages:** Python (PyTorch, Triton, Pandas), CUDA, C/C++, JavaScript, Go, Java, SQL, Matlab.

**Frameworks & development:** AWS, Databricks, Docker, Kubernetes Node, React, Django.